

Caesura User Manual

A practical guide to building probabilistic Carnatic-inspired rhythm, reading the timeline, automating musical parameters, and routing the resulting MIDI with confidence.



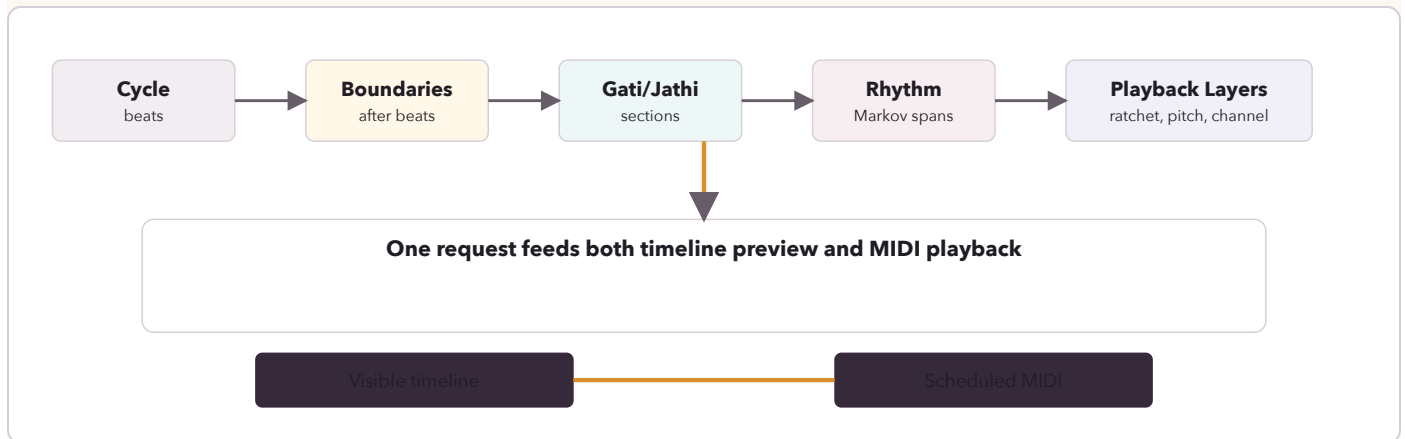
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What Caesura Is

Caesura is a macOS MIDI sequencer for Carnatic-inspired probabilistic rhythm. It builds a rhythm tree, realizes stochastic choices under seed control, shows the realization in a timeline, and sends that same realization to MIDI playback.

The core promise: what is visible in the timeline is what playback is using. If the timeline and sound disagree, treat that as a bug.



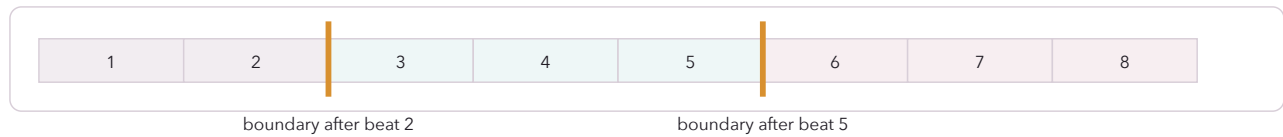
The sequencer resolves one musical request and uses it for both display and playback.

The Musical Model

A score begins with one repeating cycle. The cycle contains beats. Beats are subdivided into matras by the active gati. Section boundaries are possible changes authored after beats. Each resolved section chooses its own gati and may also choose a jathi accent pulse when the pulse tiles the section cleanly.

The most important rule: **gati subdivides each beat, never the whole section.** If a four-beat section resolves to gati 7, each beat has seven matras, for 28 matras in that section.

Cycle: 8 beats



Gati subdivides each beat
gati 4



gati 7



jathi pulse



Section start
still counts if gati repeats

Boundaries are after beats; gati subdivides every beat in the resolved section.

Five-Minute First Patch

1. Set **Beats/cycle** to 8.
2. Set **Pitch** to 60 and **Velocity** around 96.
3. Use initial gati weights for 4, 3, and 5.
4. Add boundaries after beats 2 and 5.
5. Give each boundary a moderate chance, such as 35 to 60 percent.
6. Give the boundaries their own gati weights so changes have character.
7. Set max-section weights so one or two changes are more likely than many changes.
8. Press Play and inspect the realized timeline.
9. Switch seed mode from per-cycle to locked when you find a useful variation.
10. Save the patch before adding Rhythm Shaper or ratchet.

Reading The Timeline

The timeline has authored and resolved layers. The top rail shows where boundaries may happen. The resolved bands show what actually happened for the inspected cycle. Lower lanes show gati, jathi, rhythm grouping, ratchet, ornaments, pitch, channel, and automation samples.

Authored

Boundary positions, probabilities, and weights are values you edit.

Resolved

Sections, gati, jathi, and rhythm groups are concrete choices for a cycle.

Playback

Ratchet, ornaments, pitch, and channel lanes show final MIDI behavior.

Score Controls

Control	What it changes	Notes
Beats/cycle	Cycle length.	Structural. Boundaries are normalized against it.
Pitch	Base MIDI note.	Current automation target.
Velocity	Base MIDI velocity.	Current automation target before accent sampling.
Initial gati weights	First section gati choice.	Weights are current automation targets.
Initial jathi weights	First section jathi accent choice.	Only tileable jathis can be used.
Boundaries	Possible section changes after beats.	Probability and weights are current automation targets.
Max sections	How many sections may exist.	Chosen before boundary rolls.
Single-parameter modulation	Continuity preference for rhythmic changes.	Current boolean automation target.
Accent ranges	Velocity boosts for beat, section, and jathi starts.	Min/max fields are current automation targets.

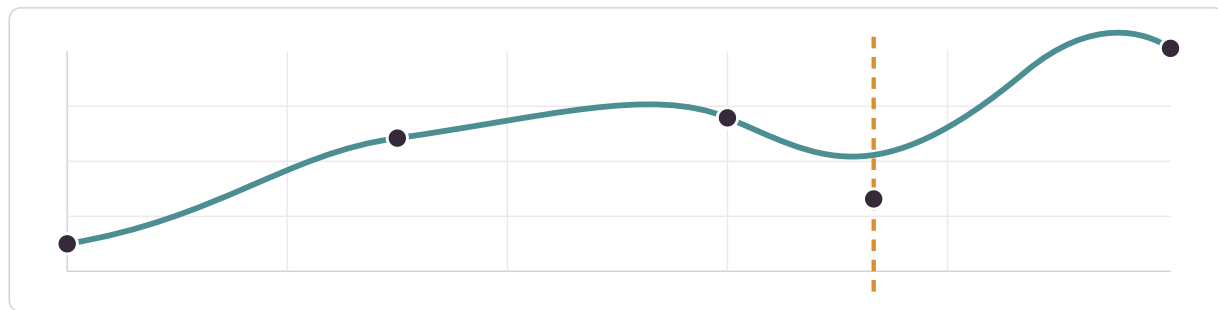
Seed Strategy

Seeds make probabilistic music repeatable. Locked mode repeats a result. Per-cycle mode changes by cycle while remaining deterministic. History mode chooses between remembered seeds and new seeds. Rhythm, pitch, and channel layers can follow the global seed or use their own strategy.

Use locked seeds to capture a phrase. Use per-cycle seeds for variation. Use history mode when a piece should revisit earlier states.

Automation

Automation runs across a top-level number of cycles. It is sampled before each beat, then the timeline and playback are realized from those samples. Points are stored as exact normalized rational times, so a point at 73 percent stays coordinated when the automation span is stretched from a short phrase to hundreds of cycles.



point list

curve amount

snap to markers

sample once per beat

Automation is graph-edited, marker-aligned, and sampled at beat boundaries.

Current Automation Targets

Current targets include pitch, velocity, single-parameter modulation, accent ranges, initial gati/jathi weights, boundary probabilities, boundary gati/jathi weights, and max-section weights. Future phases should add matrix cells, articulation cells, arbitrary subdivision, ratchet, ornaments, pitch, channel, synth monitor values, and other editable playback parameters.

Rhythm Shaper

The Rhythm Shaper groups material inside protected accent spans. A state is a pulse pattern for a span length. A matrix cell is a weighted transition from previous state context to next state. First-order mode uses one previous state; second-order mode uses two.

States



Transition Weights

0	4	1	2
1	0	6	0
3	2	0	1

Fallback

static or weighted

Seed mode

repeat or vary

Rhythm matrices choose grouped patterns inside protected spans.

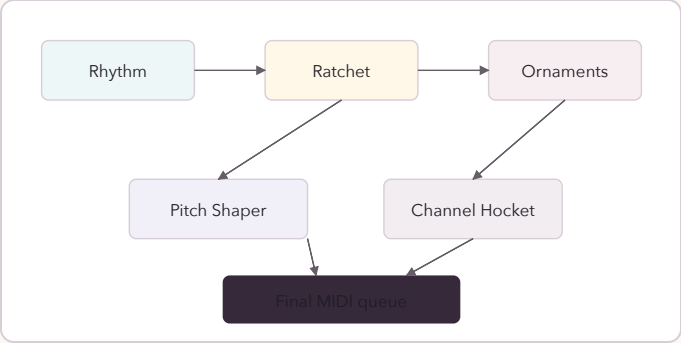
Articulation, Speed, And Re-Subdivision

Articulation can rest or tie individual cells. Gati speed and jathi timing can alter the span frame before grouping. Arbitrary subdivision can reinterpret a protected span as another virtual length, using Markov material or a weighted pattern pool. These features are powerful because they change the vocabulary the matrix sees.

Ratchet, Ornaments, Pitch, And Channel

Ratchet is a playback-only layer. It adds rapid repeated hits inside eligible note groups without changing the underlying rhythm tree. Ornaments add grace notes and delayed echoes. Pitch Shaper rewrites final pitches. Channel Hocket rewrites final MIDI channels.

These layers happen late in the pipeline, so their debug logs are important. They show what actually fired during playback.



Layer	Key controls	Debug signal
Ratchet	Probability, seed, speed windows, curves, modifiers, cooldown, velocity.	Ratchet playback events.
Ornaments	Grace placement/counts, delay windows, tuplets, shared modifiers.	Ornament playback events.
Pitch Shaper	Pitch states, Markov matrix, boundary policy, transpose, ratchet/ornament pitch modes.	Pitch playback events.
Channel Hocket	Enabled channels, Markov matrix, fallback, ratchet/ornament channel modes, accent routing.	Channel playback events.

Saving, Setup, And Logs

A `.caesura`` patch saves transport values, score controls, automation, rhythm matrices, ratchet and ornament settings, pitch shaping, channel hocketing, synth voices, setup preferences, seed paths, and relevant editor state. Autosave is crash recovery, not a replacement for deliberate patch saves.

The MIDI out log shows scheduled outgoing events. The automation playback log shows sampled automation states for each beat during playback. Seed, ratchet, ornament, pitch, and channel logs are debugging tools for proving the timeline and playback are describing the same realization.

Complete Control Reference

The developer and testing companion document is `docs/EDITABLE_VALUE_INVENTORY.md`. It lists every persisted editable value family, including dynamic matrix cells and planned automation targets, and should be updated whenever a new control is added.

Glossary

Anuloma	Speeding relationship, represented by faster speed choices.
Beat	A primary pulse inside the cycle.
Boundary	A possible section change authored after a beat.
Cycle	The repeating formal unit.
Gati	Beat subdivision count. Gati subdivides each beat.
Jathi	A regular accent pulse that must tile a resolved section.
Matra	The subdivision unit inside a beat.
Protected span	Accent-delimited region that rhythm grouping and ratchet should not cross.
Pratiloma	Slowing relationship, represented by slower speed choices.
Ratchet	Playback-only repeated hits inside note groups.
Section	A resolved region between accepted boundaries.
Seed	Integer source for reproducible stochastic choices.